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About Me

I am Deepanshu Shrivastav, currently pursuing the B.S. Degree in Data Science and Applications from IIT Madras. I enjoy building full-stack web applications and solving real-world problems using Python, Flask, Vue.js, SQLAlchemy and SQLite. I am passionate about Artificial Intelligence, Machine Learning and backend development.

Project Description

TrakoPedia is a Trekking Management Application (TMA) that helps adventure organizations manage trekking routes, trek staff, users and bookings. The application provides separate dashboards for Admin, Trek Staff and Trekkers. Admin can create treks, manage staff and monitor bookings. Trek Staff manage assigned treks and participants, while users can browse treks, book adventures and view their trekking history.

Technologies Used

1. Flask (Backend API)
2. Vue.js (Frontend)
3. Bootstrap 5 (UI)
4. SQLAlchemy (ORM)
5. SQLite (Database)
6. Flask-Security (Authentication & Roles)
7. Jinja2 (Application Entry Point)
8. Python Standard Libraries

Database Schema Design

User: Stores Admin, Trek Staff and Trekker information.
Role: Stores application roles (Admin, Staff, User).
Trek: Stores trek details including location, duration, difficulty, slots and dates.
Booking: Stores booking records for users and treks.

Relationships

- One User can book multiple treks.
- One Trek can have multiple bookings.
- Each Booking belongs to one User and one Trek.
- Admin manages treks and staff.
- Trek Staff manage assigned treks and participants.

Project Video

Paste the project demonstration video link here (maximum 3 minutes).